

Wireless Mouse Charger

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A wireless mouse is powered by a battery and does not have the option for charging. The user has to carry a spare battery to

PARTS LIST

Semiconductors:

- IC1 - LTC4054-4.2 lithium-ion battery charger
LED1 - 3mm LED

Resistors (all 1/4-watt carbon):

- R1 - 2-kilo-ohm, $\pm 1\%$
R2 - 470-ohm, $\pm 5\%$

Capacitors:

- C1, C2 - 10 μ F, 16V electrolytic

Miscellaneous:

- CON1 - 2-pin connector
- Mini-USB type B connector
CON2 - 2-pin connector
- 3.7V Li-Ion battery

replace the dead one. If a rechargeable battery is in use, provision for an adaptor for charging has to be in place. Here is a circuit built with a few components for charging a lithium-ion battery, without taking it out from the mouse. It is a constant-current charger with inbuilt auto-cut.

Circuit and working

Circuit diagram of the charger is shown in Fig. 1. It is built around LTC4054 (IC1), which is a complete constant-current/constant-voltage linear charger for lithium-ion batteries. Its thin SOT package and low external component count makes LTC4054 ideal for portable applications. It is designed specifically to

work within USB power specifications.

Charge voltage is fixed at 4.2V and charge current can be programmed externally with a single resistor. It automatically terminates

the charge cycle when the charge current drops to 1/10th of the programmed value after the final float voltage is reached.

The battery can deliver charge current up to 800mA. Programming has been done for the charging current of 500mA, which obviates the need for heat-sinks.

When IC1 determines that batteries need to be recharged, the charge pin (pin 1) automatically connects to the power line and starts charging. This status is indicated by LED1 in ON state. IC1 provides 4.2V charge voltage if the battery is in charging mode.

Resistor R1 is used to set (program) the level of charge current.



Tolerance of the resistor must be one per cent.

When batteries are almost charged, the charger enters the constant-voltage portion of the charge cycle and the charge current begins to drop. When it drops below 1/10th of the programmed current, the charge cycle ends and LED1 turns off, indicating that the charge cycle has ended.

Construction and testing

Block diagram of the circuit assembly within a mouse is shown in Fig 2. An actual-size, single-side PCB for the mouse charger circuit is shown in Fig. 3 and its component layout in Fig. 4. All SMD components such as R1, R2, C1 and C2 are to be soldered to the bottom side after proper identification.

The layout is compact and the PCB can be fitted inside the mouse. Body of the mouse needs to be modified so that a mini-USB type B PCB socket (CON1) and LED1 are visible from outside. Add multiple diodes (1N4148) to get near 3.0V, if there is any deficiency in mouse performance.

To fit the circuit inside the mouse, one has to select components based on space available inside the mouse. The

circuit was tested as per arrangement shown in Fig. 2. ●



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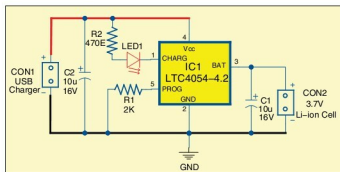


Fig. 1: Circuit diagram of the wireless mouse charger

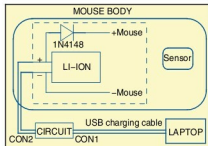


Fig. 2: Block diagram of the circuit assembly within a mouse



Fig. 3: Actual-size PCB pattern of the circuit for the wireless mouse charger



Fig. 4: Component layout of the PCB

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